

Practical 9th

Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.

CODE :

```
#include<stdio.h>
#include<stdlib.h>
int main()
{
    int request[8] = {98,183,41,122,14,124,65,67};
    int visited[8] = {0};
    int head = 53;
    int total = 0;
    int i,j;
    printf("Head Movement Sequence:\n");
    for(i=0;i<8;i++)
    {
        int index=-1;
        int min=1000;

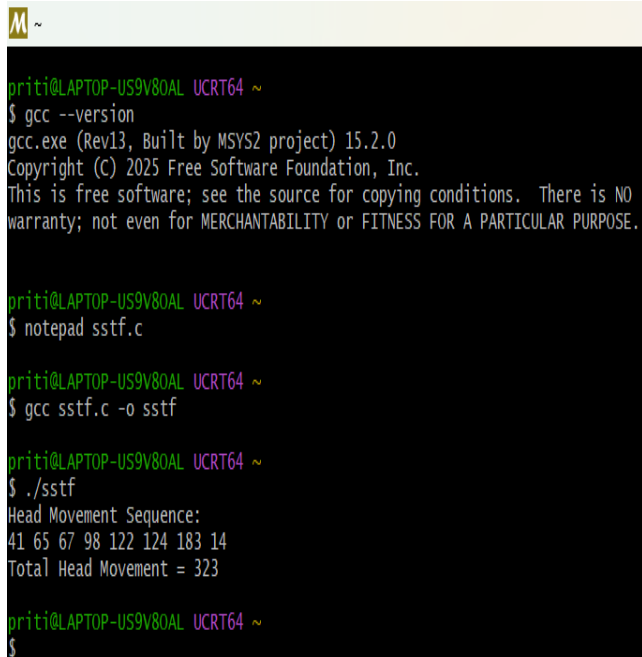
        for(j=0;j<8;j++)
        {
            if(visited[j]==0)
            {
                int distance = abs(head - request[j]);
                if(distance < min)
                {
                    min = distance;
```

```

        index = j;
    }
}
}
visited[index] = 1;
total += min;
head = request[index];
printf("%d ", head);
}
printf("\nTotal Head Movement = %d\n", total);
return 0;
}

```

OUTPUT :



```

priti@LAPTOP-US9V80AL UCRT64 ~
$ gcc --version
gcc.exe (Rev13, Built by MSYS2 project) 15.2.0
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priti@LAPTOP-US9V80AL UCRT64 ~
$ notepad sstf.c

priti@LAPTOP-US9V80AL UCRT64 ~
$ gcc sstf.c -o sstf

priti@LAPTOP-US9V80AL UCRT64 ~
$ ./sstf
Head Movement Sequence:
41 65 67 98 122 124 183 14
Total Head Movement = 323

priti@LAPTOP-US9V80AL UCRT64 ~
$

```